

The Art of Reverse Engineering



A Crackme program is an executable file and usually, on execution, you are supposed to provide a password. This article is a tutorial on cracking passwords for Crackme programs that are designed to test the skills of coders and programmers.

Wikipedia defines reverse engineering as the process by which a man-made object is deconstructed to reveal its design and architecture or to extract knowledge from the object. But what does this term mean to a software engineer? Software reverse engineering is the analysis of software to obtain information about its design and implementation. Practical applications of software reverse engineering include detecting viruses, worms, trojans and other malware, designing better software, etc. It is often believed that with source code readily available all the time, open source software systems do not need reverse engineering. But this is not true. Software reverse engineering is also done for fun and to learn. An excellent example is a Crackme program that is used to test a programmer's reverse engineering skills.

Do note that applying reverse engineering techniques on proprietary software may not be legal due to intellectual property rights. As we all know, proprietary software does not provide the source code to the user. The developers of proprietary software use code protection schemes and algorithms to conceal the code from casual disassembly. Since proprietary software is legally copyright protected, reverse engineering such software is not legal. However, small programs called Crackme programs are specifically designed for students of software

engineering, just for practising their reverse engineering skills. They are, on average, more difficult to reverse engineer than proprietary software.

Now let us look at how a Crackme program works. Usually, it is an executable file and, on execution, you are supposed to provide a password. But you are given no clues about the password and, hence, finding it by using brute force techniques alone is very difficult. In this article, we will learn a few simple Linux commands and utilities to break Crackme programs compiled in the executable format (ELF – Executable and Linkable Format) of Linux. Many online resources provide Crackme programs in the executable format (EXE) of Microsoft Windows also.

To keep the discussion short and simple, the Linux commands and tools to perform reverse engineering discussed in this article will be tested against simple C programs written by the authors themselves. Let us first consider the following C program called *crack1.c*:

```
#include<stdio.h>
#include<string.h>
#define pass "hello"
int main(
{
    char pas[100];
    printf("\nEnter the password:");
    scanf("%s",pas);
    if(!strcmp(pass,pas))
```

```
{
    printf("\nYou have CRACKED me!!!\n");
}
else
{
    printf("\nSorry\n");
}
return(0);
}
```

First, let us compile the program *crack1.c* with the command `gcc crack1.c -o test1` to obtain an executable file named *test1*. Figure 1 shows the compilation and execution of the program *crack1.c*. A typical execution of a Crackme program looks similar. The program asks the user to enter a password; it then checks whether the correct password has been entered or not, and then an appropriate message is displayed. In this case, we know the password is *hello* because we have seen the source code of the program *crack1.c*. But what if we only have the executable file *test1* of the program *crack1.c*? Then, to a person unskilled in reverse engineering, the only option is to guess the password repeatedly. But we know that such an approach is pointless because we have no idea about the number of characters in the password. This is when we need to perform reverse engineering on the executable file *test1* to make an educated guess about the password.

First, let us use a simple Linux command called *strings* to find out the